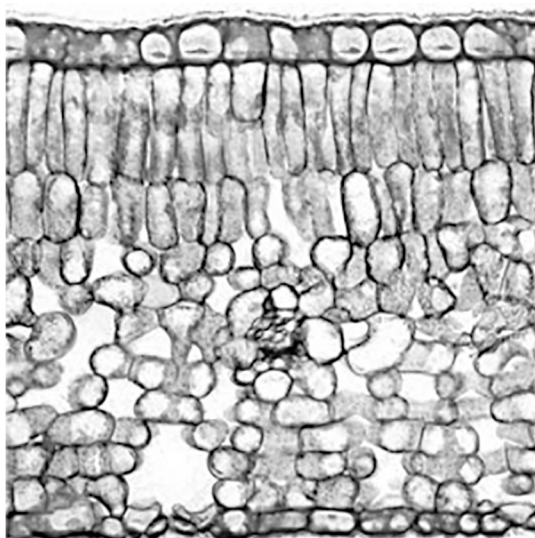


0	7
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Figure 8 is a photograph of a vertical section through a leaf observed using an optical microscope.

Figure 8



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In the box below:

- produce a scientific drawing of the arrangement of tissues shown in **Figure 8**
- label **one** of the tissues in your drawing

Do **not** draw individual cells.

[3 marks]



Question 7 continues on the next page

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**

Turn over ►



A scientist investigated factors that affect the shelf life of cut flowers.

A cut flower is the part of the stem with the flower bud attached after it has been cut from a plant.

The shelf life is the number of days the cut flowers are in good enough condition to be sold.

He:

- took 12 cut flowers from a rose plant
- determined the mean number of stomata per mm^2 on the leaves
- determined the transpiration rate for each cutting
- stored the cut flowers on a shelf in a brightly lit room
- determined the shelf life of the cut flowers.

Table 4 shows his results.

Table 4

Month cut flowers were obtained	Mean number of stomata / mm^{-2} (± 2 SD)	Mean transpiration rate / $\text{cm}^3 \text{ day}^{-1}$ (± 2 SD)	Mean shelf life / days (± 2 SD)
December	23 (± 2)	22 (± 3)	5 (± 1)
April	20 (± 3)	15 (± 2)	16 (± 2)

A value of ± 2 SD (standard deviations) from the mean includes over 95% of the data.



Using information in **Table 4**, what can you conclude about the effect of different factors on the mean shelf life of cut flowers?

[4 marks]

[illegible]

Other than a change in temperature, give **one** change the scientist could make to the environmental conditions to increase the cut flowers' shelf life.

[3 marks]

[illegible]

Turn over ►

